EXPERIMENT :-4(A)

CODE:-

```
#include <stdio.h>

struct Edge {
   int u, v, w;
};

int parent[20];

int find(int x) {
   while (parent[x] != x)
      x = parent[x];
   return x;
}

void unionSet(int a, int b) {
   parent[find(a)] = find(b);
}

int main() {
   int n, e, i, j, count = 0, cost = 0;
   struct Edge edge, temp;

   printf("Enter number of vertices and edges: ");
   scanf("%d%d", &n, &e);
```

```
struct Edge edges[e];

printf("Enter edges (u v weight):\n");
for (i = 0; i < e; i++)
  scanf("%d%d%d", &edges[i].u, &edges[i].v, &edges[i].w);

for (i = 0; i < n; i++)
  parent[i] = i;

// Sort edges
for (i = 0; i < e - 1; i++) {
  for (j = 0; j < e - i - 1; j++) {
    if (edges[j].w > edges[j + 1].w) {
      temp = edges[j];
      edges[j] = edges[j + 1];
      edges[j + 1] = temp;
    }
  }
}

printf("\nEdges in MST:\n");

for (i = 0; i < e && count < n - 1; i++) {
  edge = edges[i];

  if (find(edge.u) != find(edge.v)) {
    printf("%d - %d = %d\n", edge.u, edge.v, edge.w);
    cost += edge.w;
```

```
            unionSet(edge.u, edge.v);
            count++;
        }
    }

    printf("Minimum Cost = %d\n", cost);

    return 0;
}
```

Example

Input:

```
Enter number of vertices and edges: 4 5

Enter edges:
0 1 10
0 2 6
0 3 5
1 3 15
2 3 4
```

Kruskal first sorts:

```
2-3 = 4
0-3 = 5
0-2 = 6
0-1 = 10
1-3 = 15
```

It selects:

```
2-3 = 4
0-3 = 5
0-1 = 10
```

So:

```
Minimum Cost = 19
```